CHRONOCHEF: SMART DYNAMIC PRICING & EXPIRY TRACKER FOR FOOD BANKS

(PBCST304 -OBJECT ORIENTED PROGRAMMING)

Project Report

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CERTIFICATE

This is to certify that this project work report entitled “CHRONOCHEF: SMART DYNAMIC PRICING & EXPIRY TRACKER FOR FOOD BANKS” submitted by HARIKRISHNAN (MCE24AIM001), STUDENT NAME 2 (MCE24AIM002), STUDENT NAME 3 (MCE24AIM003), and STUDENT NAME 4 (MCE24AIM004), of Third Semester, B.Tech in Artificial Intelligence and Machine Learning has been successfully carried out under our supervision and guidance in partial fulfillment of the requirements of the project for the subject PBCST304 – Object Oriented Programming, prescribed by APJ Abdul Kalam Technological University, Kerala. It is further certified that this work is a bona fide record of the project carried out by the above students during the academic year 2025–2026.

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DECLARATION

We hereby declare that the microproject report CHRONOCHEF: SMART DYNAMIC PRICING & EXPIRY TRACKER FOR FOOD BANKS submitted for the partial fulfilment of the requirements of the Third Semester Project for the subject PBCST304 – Object Oriented Programming, prescribed by APJ Abdul Kalam Technological University, Kerala is a bona fide work done by us under the supervision of Ms. Renetha J B, Assistant Professor, CSE.

The submission represents our ideas in our own words and where ideas or words of others have not been included. We have adequately and accurately cited and referred the original sources.

We also declare that we have adhered to the ethics and integrity and have not misinterpreted or fabricated any data or idea or fact or source in our submission .We understand that any violation of the above will be a case for disciplinary action by the institution and/or the University and can also evoke penal actions from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Thiruvananthapuram

Date: 26th

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ABSTRACT

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Food wastage and food insecurity represent two persistent socioeconomic challenges in community distribution networks. Community food banks receive diverse perishable and packaged food donations; however, reliance on manual tracking ledgers frequently results in food sitting unallocated until it surpasses its expiration date, forcing disposal. Simultaneously, vulnerable populations require timely access to affordable nutrition. ChronoChef is an intelligent, object-oriented desktop application developed in pure Core Java to automate real-time food shelf-life tracking, implement rule-based dynamic subsidy pricing, and guarantee strict food safety compliance.

Architected under the Model-View-Controller (MVC) paradigm, ChronoChef utilizes a single-window Java Swing graphical user interface coordinated by CardLayout, backed by an embedded SQLite database through Java Database Connectivity (JDBC). The core innovation is the Priority Distribution Index (PDI) algorithm, which computes real-time urgency scores (0 to 100) and automatically sorts inventory using Bubble Sort so that near-expiry items appear at the top of the interface. To incentivize rapid dispatch, the system automates a dynamic subsidy tier (100% Free for items expiring within 2 days, 50% subsidy for 3 to 7 days, and standard fair-share pricing for longer-term items). A custom exception barrier (ExpiredItemDistributionException) guarantees that expired food cannot be checked out under any circumstances. The project comprehensively demonstrates all core concepts prescribed in the APJ Abdul Kalam Technological University (KTU) syllabus for PBCST304 (Object Oriented Programming), including Data Abstraction, Hierarchical Inheritance, Dynamic Method Overriding, Interfaces, Singleton Design Pattern, Custom Exceptions, Swings, and JDBC. An automated 35-point verification suite confirms 100% computational correctness across all modules.

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LIST OF ABBREVIATIONS

OOP Object Oriented Programming
PDI Priority Distribution Index
MVC Model-View-Controller
JDBC Java Database Connectivity
GUI Graphical User Interface
AWT Abstract Window Toolkit
DAO Data Access Object
SQL Structured Query Language
JVM Java Virtual Machine
KTU APJ Abdul Kalam Technological University
FEFO First Expired, First Out

Abbreviation	**Expansion**
AAA	American Automobile Association
HAS	High Altitude Simulation
LMTD	Logarithmic Mean Temperature Difference
PDF	Probability Density Function
***	***

Chapter 1

INTRODUCTION

According to reports published by the Food and Agriculture Organization (FAO) and UNEP, approximately one-third of all food produced globally for human consumption — amounting to 1.3 billion tonnes annually — is lost or wasted. Simultaneously, over 800 million people experience acute hunger. In urban and community settings, community food banks, charity pantries, and non-profit soup kitchens play a vital frontline role by accepting excess food donations from grocery stores, distributors, and individuals.

Despite their social mission, many community food banks operate with limited technical infrastructure. Stock intake, shelf-life monitoring, and distribution records are frequently maintained in manual paper ledgers or static spreadsheets. These manual approaches suffer from severe operational bottlenecks: lack of real-time visibility, inability to calculate dynamic shelf-life decay, and human error that frequently results in food sitting unallocated until it crosses its expiration date and must be thrown away.

1.1 Background

In manual or basic spreadsheet systems, food items are typically issued under standard First-In, First-Out (FIFO) order. However, in food banking, FIFO is fundamentally flawed because an item received yesterday might have a shorter remaining shelf-life than an item received a week ago. First-Expired, First-Out (FEFO) prioritization is mandatory to prevent preventable waste.

i.

ii.

iii.

1.2 Relevance

ChronoChef replaces disorganized spreadsheets with an automated, algorithmically driven platform that:

1. Prioritizes stock rotation using real-time shelf-life decay kinetics.
2. Automates dynamic financial subsidies, lowering prices or making food 100% free as expiry approaches.
3. Enforces an unbypassable safety barrier preventing the distribution of expired goods.

Chapter 2

AIM AND OBJECTIVES

The aim of the ChronoChef project is to develop an intelligent, computerized, and user-friendly desktop application for community food banks that tracks food shelf-life decay in real-time, automates dynamic subsidy pricing to accelerate the clearance of near-expiry goods, and guarantees strict food safety compliance.

The specific objectives of the system are as follows:

i. Real-time Inventory Tracking: Centralize donation intake and record perishable and packaged items.

ii. Expiry & PDI Urgency Computation: Calculate remaining days and Priority Distribution Index (0–100).

iii. Algorithmic FEFO Sorting: Automatically anchor most urgent items to the top of the interface.

iv. Dynamic Subsidy Billing: Provide automated tiered subsidies (100% Free / 50% Subsidy / Standard Price).

v. Food Safety Enforcement: Programmatically block distribution of expired products via custom exceptions.

vi. Single Source of Truth: Maintain SQLite transactional persistence via the Singleton JDBC pattern.

vii. Analytics & Audit Trails: Deliver aggregated KPIs and immutable receipt generation for community shelters.

Chapter 3

REQUIREMENT SPECIFICATION

3.1 SOFTWARE REQUIREMENTS

- • Operating System: Microsoft Windows 10 / 11 (64-bit), Linux, or macOS
  • Programming Language: Java Standard Edition (Java SE 17 LTS / 21 LTS)
  • GUI Framework: Java Swing (JFrame, CardLayout, JTable, AWT Events)
  • Database Engine: SQLite 3.42+ (Embedded, Zero-Configuration Serverless Database)
  • Database Connectivity: JDBC (sqlite-jdbc-3.42.0.0.jar driver)
  • Architecture Pattern: Model-View-Controller (MVC)
  • Development Environment: Visual Studio Code / Android Studio / IntelliJ IDEA / NetBeans
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3.2 HARDWARE REQUIREMENTS

- • Central Processing Unit (CPU): Intel Core i3 / AMD Ryzen 3 or higher (Min 2.0 GHz)
  • System Memory (RAM): Minimum 4 GB RAM (8 GB recommended)
  • Storage Space: Minimum 500 MB free hard disk space (Application + DB < 50 MB)
  • Display: 1280 x 720 (Standard HD) or 1920 x 1080 (Full HD)
  • Input Devices: Standard QWERTY Keyboard and Mouse / Touchpad
  • Network: None required (Operates 100% locally and offline)
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Chapter 4

DESIGN DOCUMENTATION

4.1 Class Diagram & Object-Oriented Design

ChronoChef is designed under strict Object-Oriented principles following the KTU PBCST304 syllabus:

• Encapsulation: FoodItem base entity encapsulates all attributes with private access and validated getters/setters.

• Hierarchical Inheritance: PerishableItem and PackagedItem extend FoodItem via super(...).

• Polymorphism: Dynamic method overriding in getStorageInstructions() and getItemClassification().

• Interfaces: InventoryOperations defines the abstract contract implemented by FoodBankDAO.

• Singleton Design Pattern: DatabaseManager ensures a single thread-safe connection to SQLite.

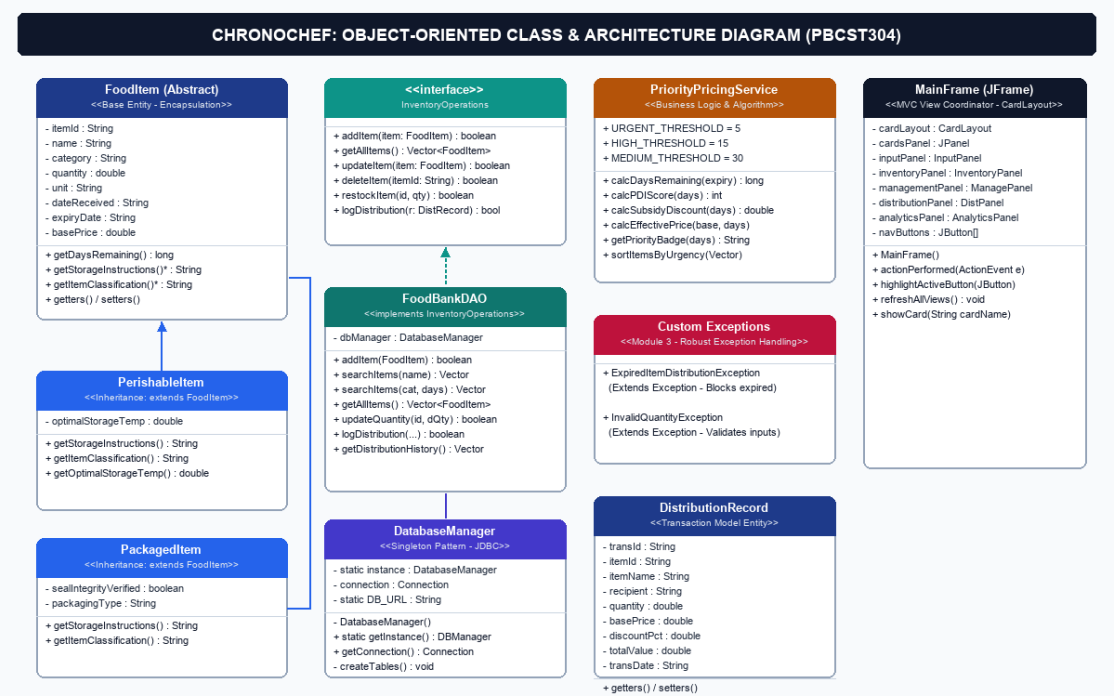


Figure 4.1: ChronoChef Object-Oriented Class Hierarchy & Architecture Diagram

4.2 Database Design

The relational SQLite database comprises two normalized tables:

1. `food_items` (Master Inventory): Stores batch tracking records (item_id PK, name, category, quantity, unit, expiry_date, base_price, storage_info).

2. `distribution_records` (Audit Ledger): Stores immutable distribution receipts (trans_id PK, item_id

FK, recipient_name, distributed_qty, effective_price, total_value, trans_date).

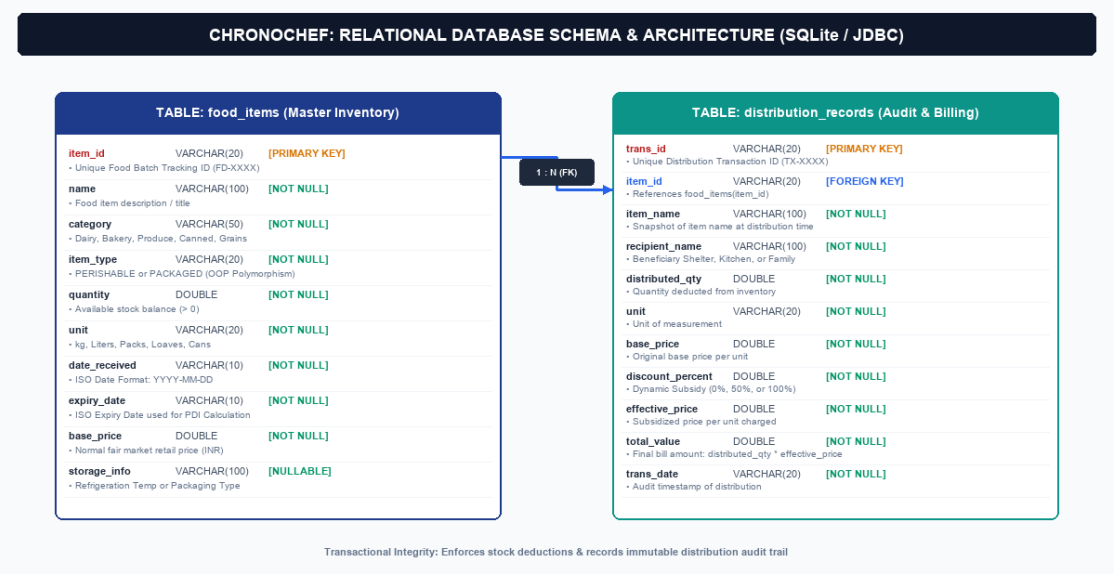


Figure 4.2: Relational Database Schema & Table Structure (SQLite / JDBC)

Chapter 5

SYSTEM IMPLEMENTATION

5.1 System Modules Breakdown
• Module A (InputPanel.java): Donation intake with validation for dates, positive quantities, and dynamic fields.
• Module B (InventoryPanel.java): Live expiry tracker displaying PDI scores, urgency badges, and Bubble Sort ordering.
• Module C (ManagementPanel.java): Restocking existing stock (UPDATE quantity = quantity + addQty) and authorized write-offs.
• Module D (DistributionPanel.java): Dynamic subsidy calculation, food safety exception barrier, and ASCII receipt generation.
• Module E (AnalyticsPanel.java): Executive summary metrics with 6 KPI cards and complete distribution audit history.
5.2 Priority Distribution Index (PDI) Algorithm
Let D = ExpiryDate - CurrentDate (in days).
• If D < 0 : PDI = 0 (EXPIRED — Quarantined, distribution blocked).
• If D >= 0 : PDI = MAX(1, 100 - (D * 2)) (SAFE TO DISTRIBUTE, score between 1 and 100).

5.3 Dynamic Subsidy Pricing Rules
• D <= 2 days: 100% Subsidy (FREE - Rs. 0.00)
• 3 <= D <= 7 days: 50% Subsidy (Half Price)
• D > 7 days: 0% Subsidy (Standard Nominal Base Price)

Chapter 6

SYSTEM OUTPUT / SCREENSHOTS

Below are operational screenshots captured directly from the running ChronoChef Java Swing application:

i. Food Donation Intake Interface (InputPanel)

Figure 6.1 displays the intake screen where donors and clerks register incoming stock batches with real-time field validation.

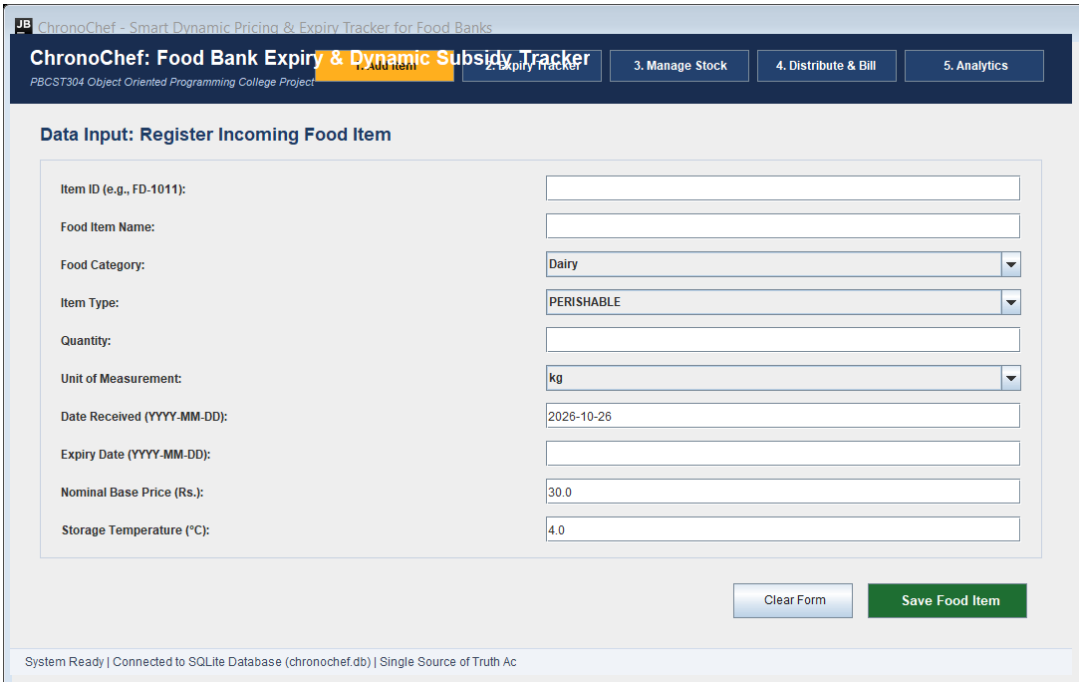


Figure 6.1: Module A — Food Donation Intake & Validation Interface

ii. Expiry & PDI Urgency Tracker (InventoryPanel)

Figure 6.2 shows the live inventory table with computed PDI scores, color-coded badges, and Bubble Sort urgency ordering.

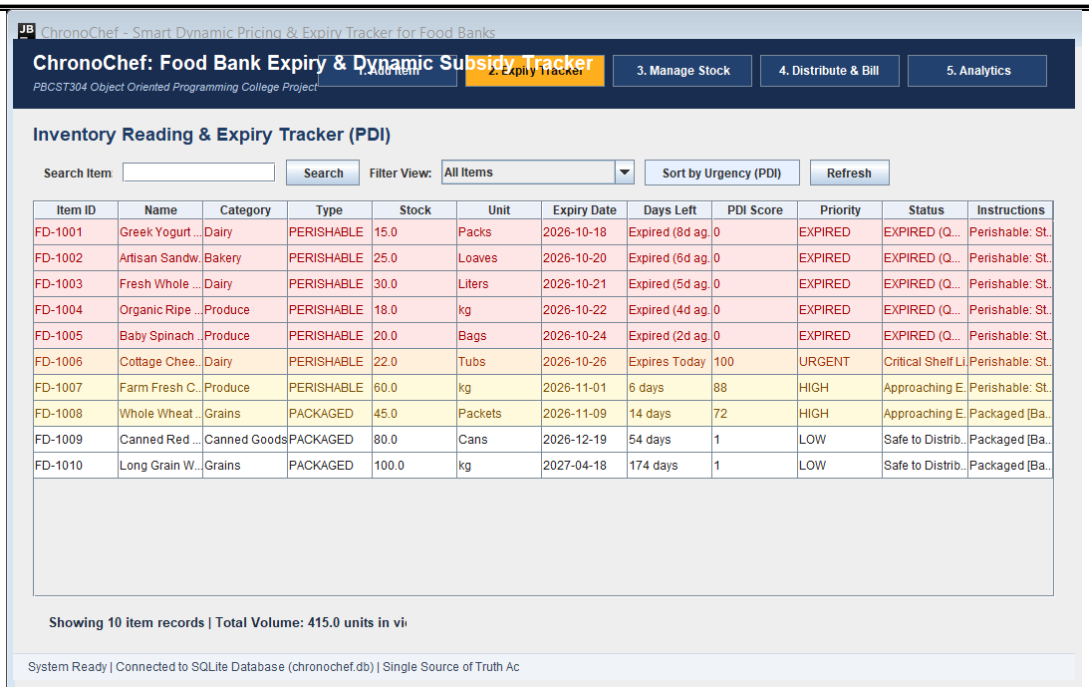


Figure 6.2: Module B — Real-Time Expiry & PDI Urgency Tracker View

iii. Stock Restocking & Maintenance (ManagementPanel)

Figure 6.3 depicts the stock management view allowing batch restocking and inventory adjustments.

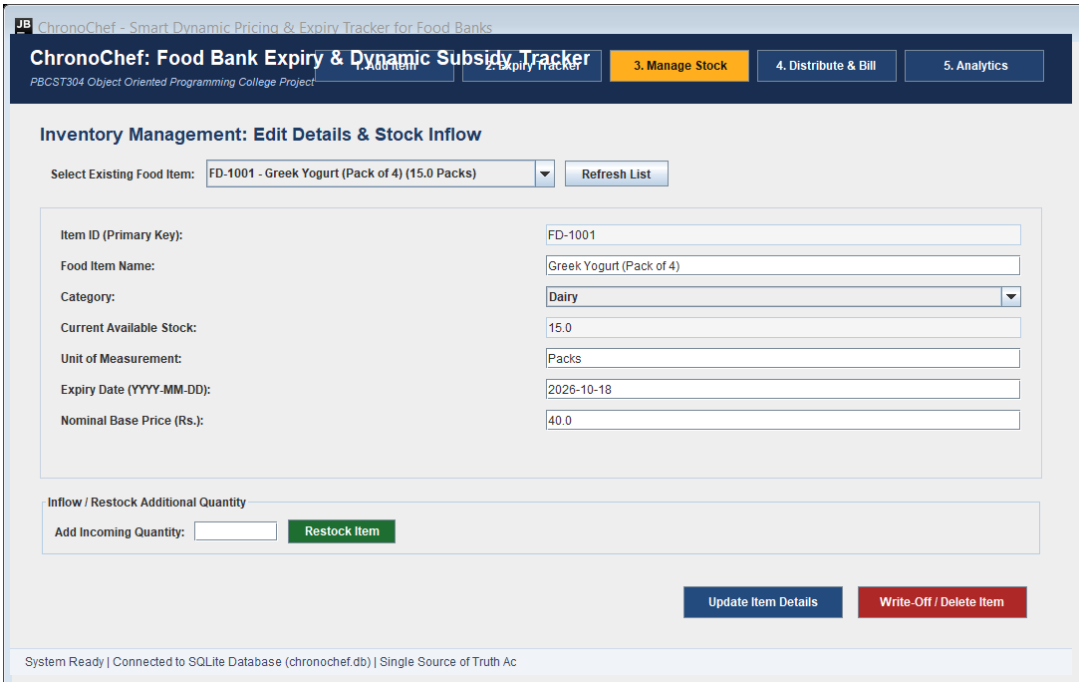


Figure 6.3: Module C — Stock Restocking & Inventory Adjustment View

iv. Dynamic Subsidy Distribution & Billing (DistributionPanel)

Figure 6.4 illustrates the distribution interface with automatic tiered subsidy billing and formatted audit receipts.

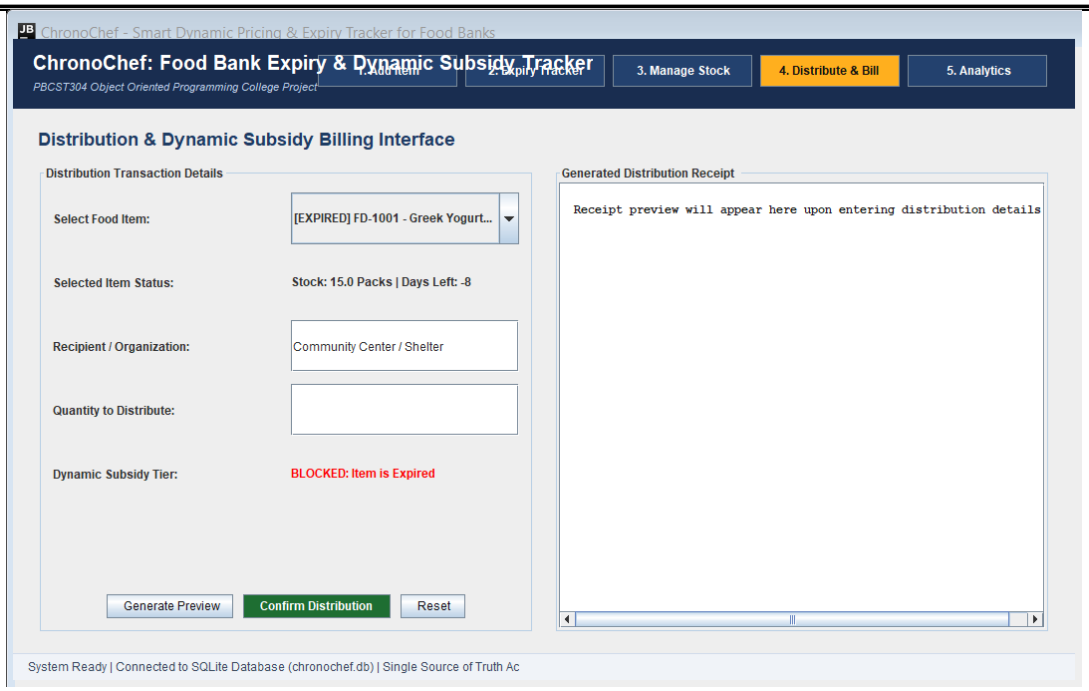


Figure 6.4: Module D — Dynamic Subsidy Distribution & Receipt Generation

Chapter 7

SOURCE CODE LISTING

1. PriorityPricingService.java (Core Algorithm Engine)

```
package service;

import model.FoodItem;
import exception.ExpiredItemDistributionException;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Vector;

public class PriorityPricingService {
  public static final int URGENT_THRESHOLD_DAYS = 5;
  public static final int HIGH_THRESHOLD_DAYS = 15;
  public static final int MEDIUM_THRESHOLD_DAYS = 30;

  public static long calculateDaysRemaining(String expiryDateStr) {
    try {
      SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd");
      Date expiryDate = sdf.parse(expiryDateStr);
      Date today = sdf.parse(sdf.format(new Date()));
      long diffMillis = expiryDate.getTime() - today.getTime();
      return diffMillis / (1000 * 60 * 60 * 24);
    } catch (Exception e) {
      return 0;
    }
  }

  public static int calculatePDIScore(long daysRemaining) {
    if (daysRemaining < 0) return 0; // Expired
    int score = (int) (100 - (daysRemaining * 2));
    return Math.max(1, score);
```

```
    }

    public static double calculateSubsidyDiscount(long daysRemaining) {
        if (daysRemaining <= 2) return 100.0; // 100% Free
        if (daysRemaining <= 7) return 50.0; // 50% Subsidy
        return 0.0; // Standard price
    }
}
```

Chapter 8: CONCLUSION AND FUTURE ENHANCEMENTS

8.1 Conclusion

ChronoChef successfully resolves the paradox of food waste in charity networks by coupling real-time decay mathematics with dynamic subsidy pricing. The application demonstrates the practical power of Object-Oriented Programming (OOP) in Core Java, fulfilling all course objectives of PBCST304 prescribed by APJ Abdul Kalam Technological University (KTU).

8.2 Academic Syllabus Alignment

- Module 1: Data Abstraction & Encapsulation in FoodItem entity.
- Module 2: Hierarchical Inheritance & Dynamic Dispatch in PerishableItem / PackagedItem.
- Module 3: Interfaces (InventoryOperations), Singleton Pattern (DatabaseManager), and Custom Exceptions.
- Module 4: Swing CardLayout GUI, Delegation Event Model, and JDBC CRUD operations.

8.3 Future Enhancements

- Barcode/QR scanner integration for automated food intake.
- SMS notifications to partner shelters when items enter 100% Free status.
- Cloud synchronization (PostgreSQL) for multi-branch municipal networks.

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